

Introduction to Electronics Engineering

MODULE 2 — PART 1: COMPLETE NOTES

Module 2 (10 hrs): Bipolar Junction Transistors (BJT) & Junction Field Effect Transistors (JFET)

For First-Year Students — written in simple, easy language
Course: Introduction to Electronics Engineering (IEE) · Course Code: EC 101

Module 2 — what it covers (as per the syllabus)

- PNP and NPN transistors — construction and characteristics of CB, CE and CC configurations
- DC and AC load line analysis, operating point
- Transistor biasing: fixed bias, emitter bias / self-bias
- Junction Field Effect Transistors: construction, characteristics, operations

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1	Introduction to the Bipolar Junction Transistor (BJT)
2	Working Principles of NPN and PNP Transistors
3	BJT Configurations (CB, CE, CC)
4	Input and Output Characteristics
5	Current Gains and Leakage Currents (with solved numericals)
6	BJT Biasing and Load-Line Analysis
7	Junction Field Effect Transistor (JFET)
8	Quick Reference — Formulae and Definitions

1. Introduction to the Bipolar Junction Transistor (BJT)

1.1 What is a Transistor?

- Besides diodes, the most popular semiconductor device is the transistor, e.g., the Bipolar Junction Transistor (BJT).
- The two most important applications of a transistor are: **as an amplifier** and **as a switch**.
- A transistor is a **three-terminal device**: Base (B), Emitter (E) and Collector (C).
- In a bipolar transistor, current conduction takes place due to **both** types of charge carriers — electrons and holes. Hence the name **Bipolar** Junction Transistor.
- In a BJT the output current is controlled by the input current, so it is a **current-controlled device**.

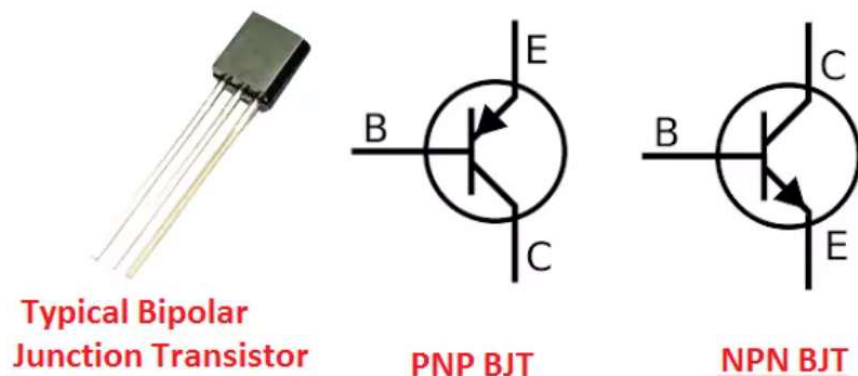


Figure 1: The Bipolar Junction Transistor — terminals, symbols and structure.

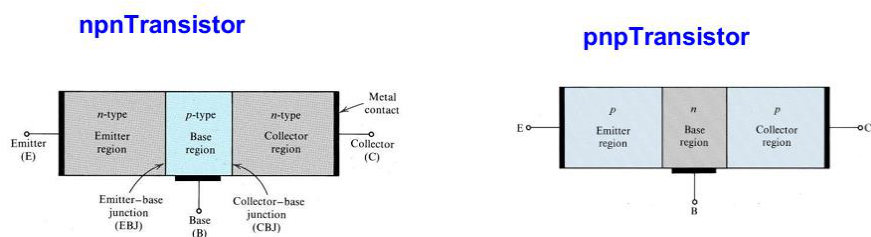


Figure 2: Internal structure of NPN and PNP transistors.

1.2 Types of BJT and Doping Levels

- NPN transistor** — emitter and collector are n-doped; base is p-doped.
- PNP transistor** — emitter and collector are p-doped; base is n-doped.
- Emitter**: heavily doped — supplies majority carriers.
- Base**: lightly doped and very thin — enables carrier transit.
- Collector**: moderately doped — collects majority carriers from the emitter.

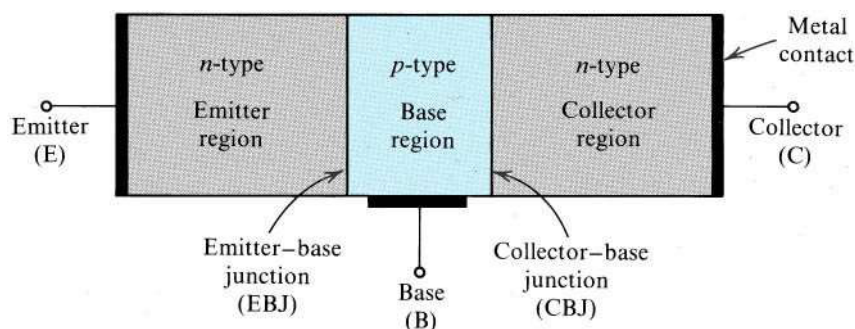


Figure 3: Cross-section of an NPN transistor.

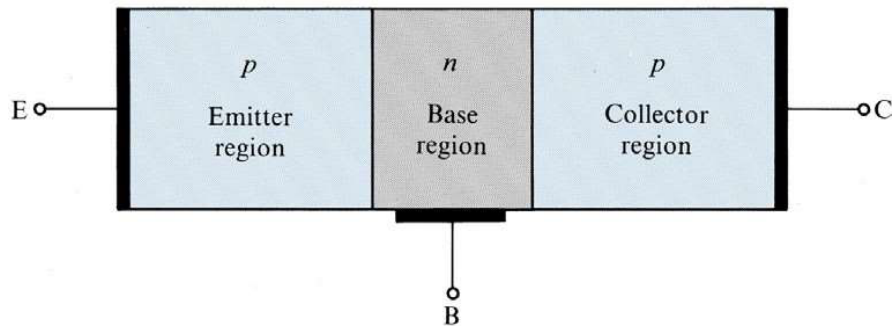


Figure 4: Cross-section of a PNP transistor.

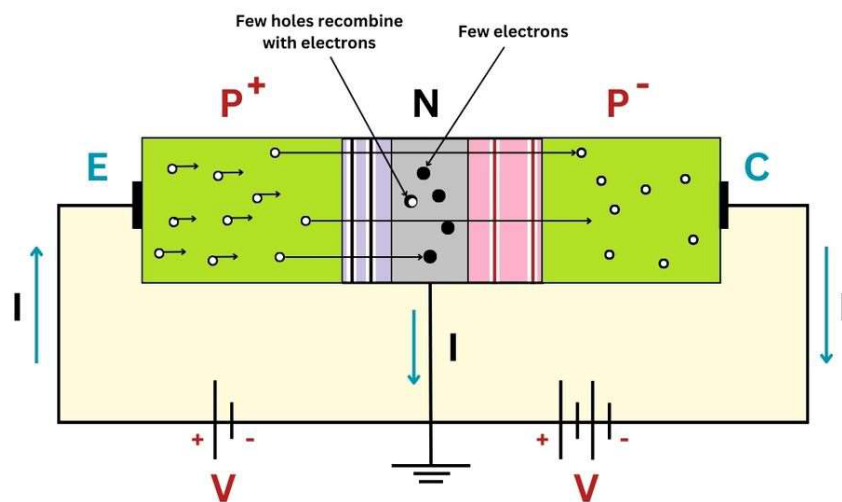


Figure 5: The BJT structure — two PN junctions (emitter-base and collector-base) share one base region.

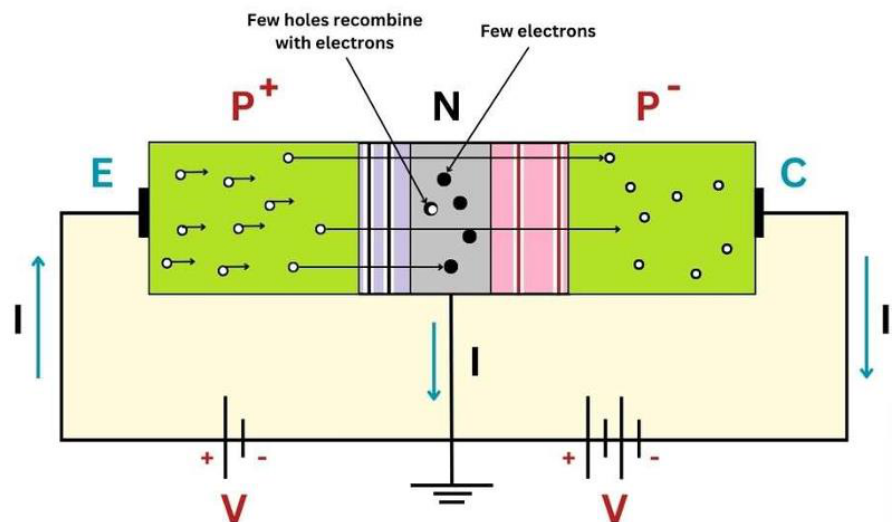


Figure 6: Internal structure of the BJT showing the bias applied to the two junctions.

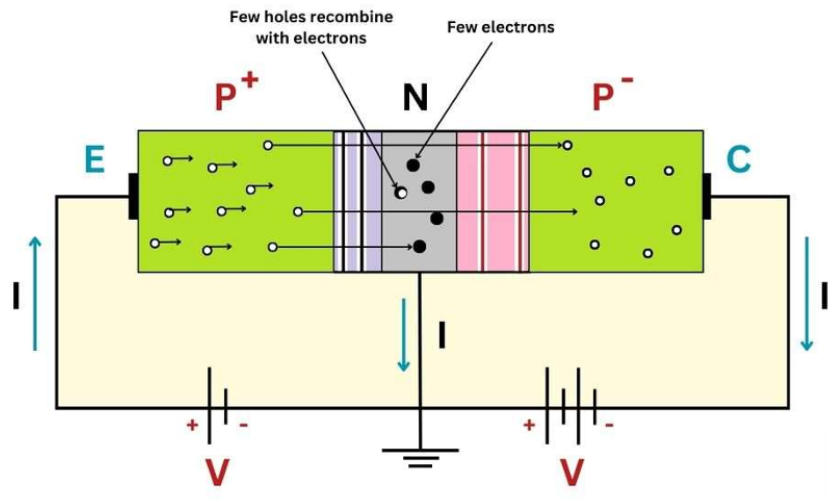
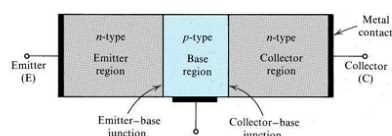


Figure 7: Charge carriers (electrons and holes) inside the transistor regions.

Dipolar Junction Transistor (BJT)

nnpTransistor



pnpTransistor

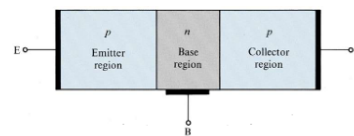


Figure 8: Current flow block diagrams for NPN and PNP transistors.

1.3 Modes (Regions) of Operation of the BJT

To operate the transistor properly as an amplifier, the two PN junctions must be correctly biased with external voltages. Depending on the external bias voltage polarities applied to the two junctions, the transistor works in one of three regions:

Region	Emitter-Base Junction (J_E)	Collector-Base Junction (J_C)	Application / Behaviour
Cut-off	Reverse biased	Reverse biased	Transistor is OFF — acts as an open switch
Active	Forward biased	Reverse biased	Amplification region — linear operation
Saturation	Forward biased	Forward biased	Transistor is ON — acts as a closed switch

2. Working Principles of NPN and PNP Transistors

2.1 NPN Transistor Operation

For proper amplifier operation, the Emitter-Base (EB) junction is forward biased and the Collector-Base (CB) junction is reverse biased — this is the **active region**.

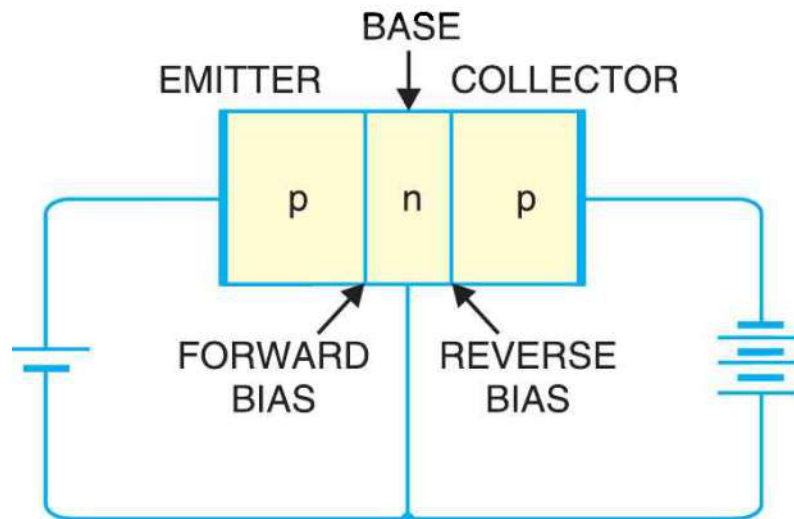


Figure 9: NPN transistor with the EB junction forward biased and the CB junction reverse biased.

- The forward-biased EB junction causes the electrons in the n-type emitter to flow toward the base — this constitutes the **emitter current** I_E .
- As these electrons flow through the thin, lightly doped p-type base, very few recombine with holes. This small recombination constitutes the **base current** I_B .
- The remaining large number of electrons cross the base region and move through the collector to the positive terminal of the external DC source — this constitutes the **collector current** I_C .
- Since most of the electrons from the emitter reach the collector and very few recombine in the base, the collector current is much larger than the base current.

$$I_E = I_B + I_C$$

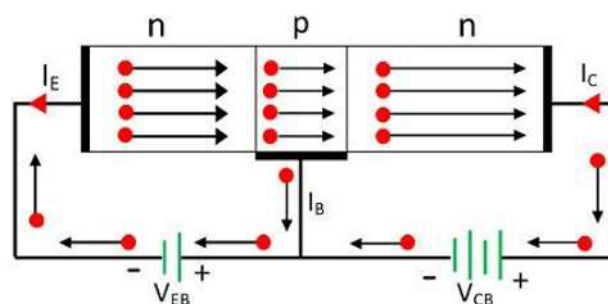


Figure 10: Current flow in the NPN transistor.

2.2 PNP Transistor Operation

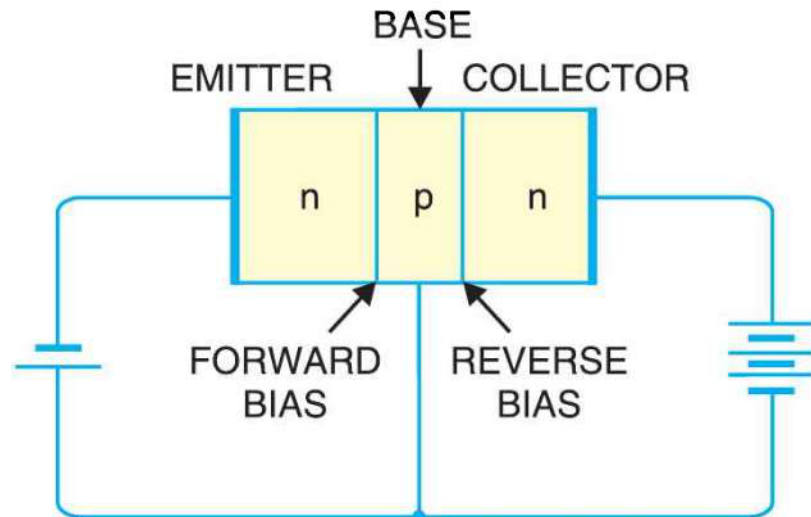


Figure 11: PNP transistor with the EB junction forward biased and the CB junction reverse biased.

- The forward-biased EB junction causes the holes in the p-type emitter to flow toward the base — this constitutes the emitter current I_E .
- As these holes flow through the thin, lightly doped n-type base, very few recombine with electrons — this small recombination constitutes the base current I_B .
- The remaining large number of holes cross the depletion region and move through the collector to the negative terminal of the external DC source — this constitutes the collector current I_C .
- Hole flow is the dominant conduction mechanism in a PNP transistor.

$$I_E = I_B + I_C$$

3. BJT Configurations

3.1 The Three Configurations

A BJT can be connected in three different configurations, each with unique characteristics:

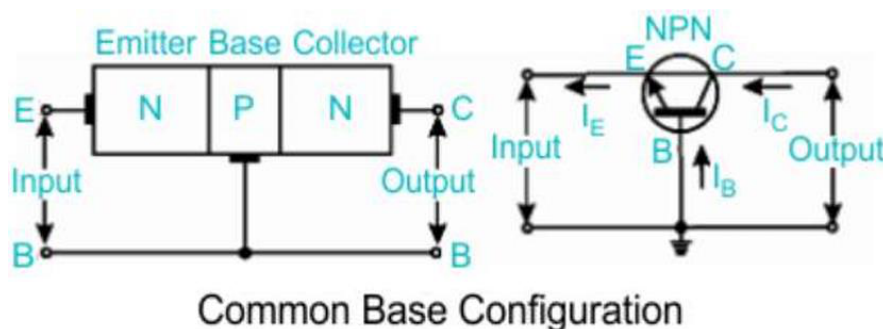


Figure 12: Common Base (CB) configuration — the base is common to input and output.

- **Common Base (CB):** the base terminal is common to both input and output circuits. Input: emitter. Output: collector. High voltage gain, low input impedance, current gain less than 1 (alpha). Used in high-frequency applications.

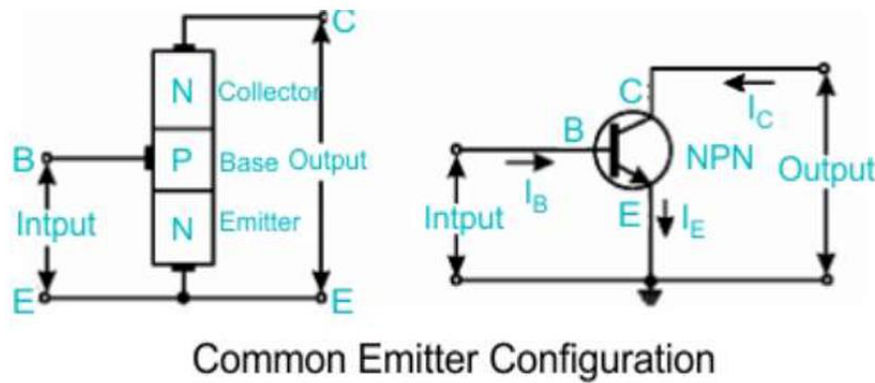


Figure 13: Common Emitter (CE) configuration — the emitter is common to input and output.

- **Common Emitter (CE):** the emitter terminal is common to both input and output circuits. Input: base. Output: collector. High voltage gain and high current gain (beta). The output is 180° phase-inverted relative to the input. This is the most widely used configuration for amplification.

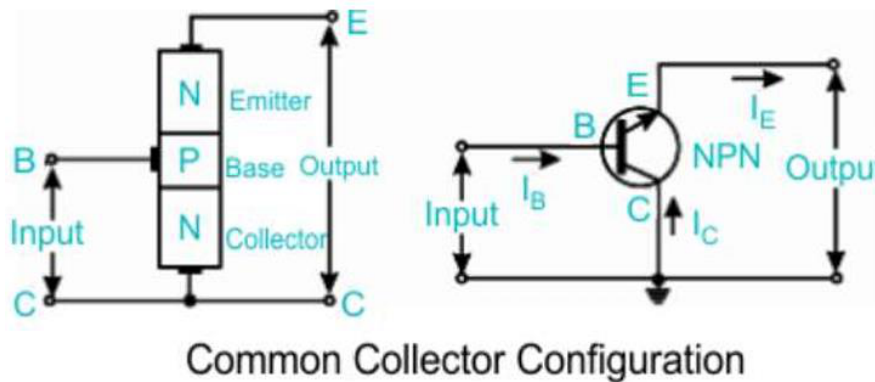


Figure 14: Common Collector (CC) configuration — the collector is common to input and output.

- **Common Collector (CC), also called Emitter Follower:** the collector terminal is common to both input and output circuits. Input: base. Output: emitter. High input impedance, low output impedance, approximately unity voltage gain. Used primarily for impedance matching and as a buffer stage.

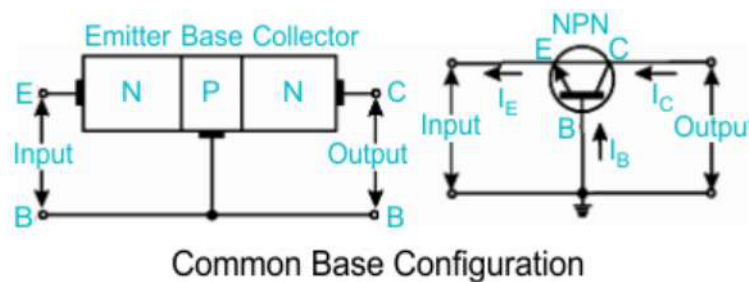


Figure 15: Circuit of a transistor connected in the common base configuration.

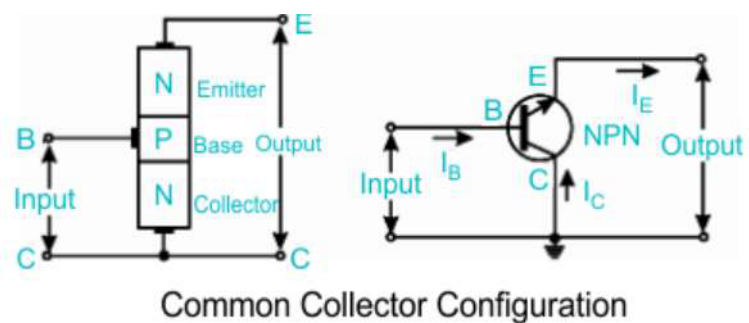


Figure 16: Circuit of a transistor connected in the common collector configuration.

3.2 Configuration Comparison Table

Parameter	Common Base (CB)	Common Emitter (CE)	Common Collector (CC)
Common terminal	Base	Emitter	Collector
Input terminal	Emitter	Base	Base
Output terminal	Collector	Collector	Emitter
Current gain	Less than 1 (alpha, typically 0.95–0.99)	High (beta, typically 20–500)	High (beta + 1)
Voltage gain	High	High	About 1 (unity)
Input impedance	Low (about 50 ohm to 0.5 k-ohm)	Medium (about 1 k-ohm)	High (about 100 k-ohm)
Output impedance	High	Medium	Low
Phase shift (voltage)	0° (no inversion)	180° (inverted)	0° (no inversion)
Primary application	High-frequency circuits	General amplification	Impedance matching / buffer

4. Input and Output Characteristics

4.1 Common Base (CB) Configuration Characteristics

Input characteristics: the variation of emitter current (I_E) with the base-emitter voltage (V_{BE}), keeping the collector-base voltage (V_{CB}) constant.

Output characteristics: the variation of collector current (I_C) with the collector-base voltage (V_{CB}), keeping the emitter current (I_E) constant.

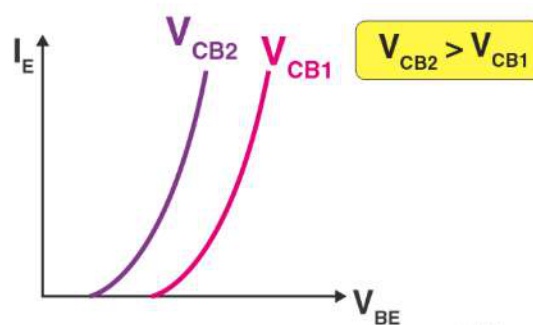


Figure 17: Input characteristics of the CB configuration.

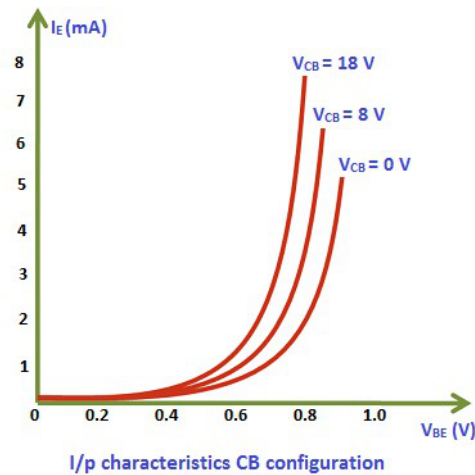


Figure 18: Input characteristic curves of the CB configuration at different values of V_{CB} .

Since the emitter-base junction is forward biased, an increase in V_{BE} reduces the barrier potential of the junction and allows more charge carriers to flow from the emitter into the base. The emitter current therefore increases rapidly with a small increase in V_{BE} , and the input curve resembles the forward-bias characteristic of a PN junction diode.

Early Effect (Base-Width Modulation)

- When the reverse bias voltage V_{CB} increases, the width of the depletion region also increases, which reduces the electrical (effective) base width.
- This reduces recombination in the base, so a slightly smaller emitter-base voltage is required to produce the same emitter current.
- As a result, the input characteristic curves shift slightly with changes in V_{CB} .

Output Characteristics of the CB Configuration

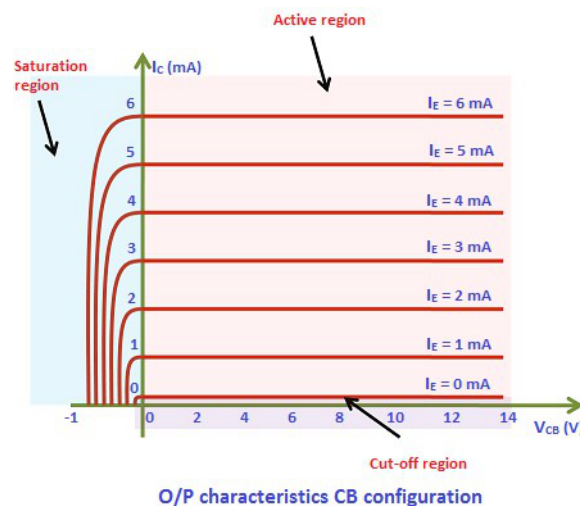


Figure 19: Output characteristics of the CB configuration.

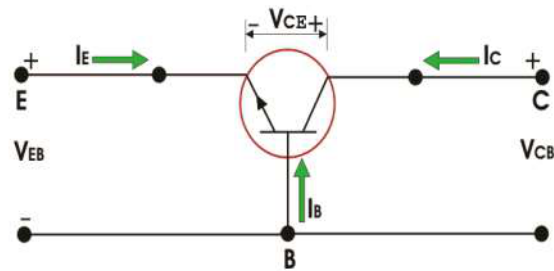


Figure 20: Output characteristic curve with the three regions marked.

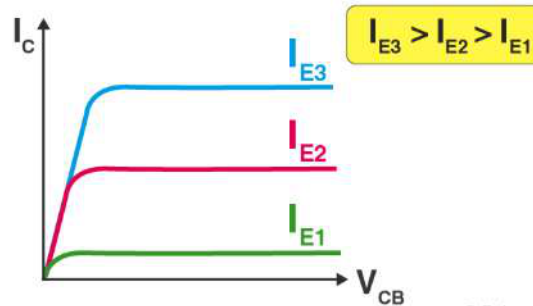


Figure 21: Output characteristics for different emitter-current values.

$$I_E = I_C + I_B$$

$$\alpha = \frac{I_C}{I_E}, \quad \beta = \frac{I_C}{I_B}$$

$$\Rightarrow \beta = \frac{I_C}{I_E - I_C} = \frac{\alpha I_E}{I_E(1 - \alpha)}$$

$$\beta = \frac{\alpha}{1 - \alpha}$$

Figure 22: CB output characteristics — active, saturation and cut-off regions.

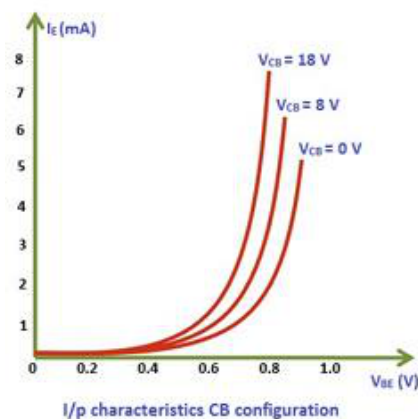


Figure 23: CB output characteristics.

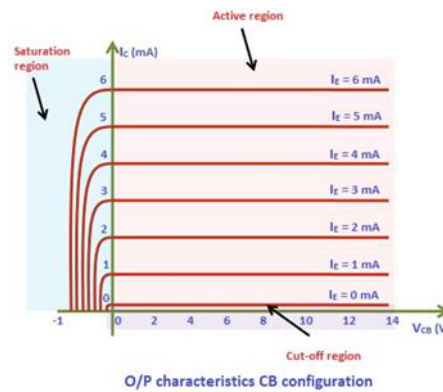


Figure 24: CB output characteristics with the regions marked.

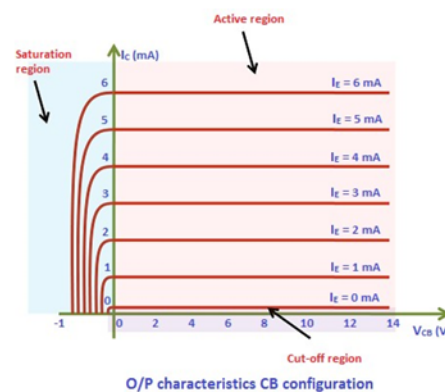


Figure 25: CB output characteristic curves.

The output characteristic curve is divided into three important regions:

- **Active region** — the region between the vertical dashed line and the horizontal axis. The collector current is approximately constant and nearly equal to the emitter current. The transistor works as an amplifier here.
- **Saturation region** — the region to the left of the vertical dashed line. In this region V_{CE} is negative for an NPN transistor, and a small change in V_{CE} produces a large change in collector current. Both junctions are forward biased.
- **Cut-off region** — the region along the horizontal axis (the curve marked $I_E = 0$). A small leakage current (I_{CBO}) still flows due to minority carriers even when $I_E = 0$.

4.2 Common Emitter (CE) Configuration Characteristics

Input characteristics: the variation of base current (I_B) with the base-emitter voltage (V_{BE}), keeping the collector-emitter voltage (V_{CE}) constant.

Output characteristics: the variation of collector current (I_C) with the collector-emitter voltage (V_{CE}), keeping the base current (I_B) constant.

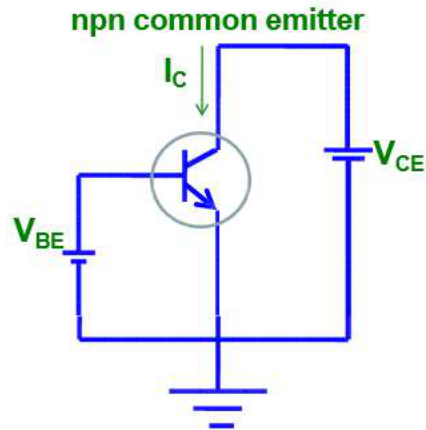


Figure 26: NPN transistor connected in the common emitter configuration.

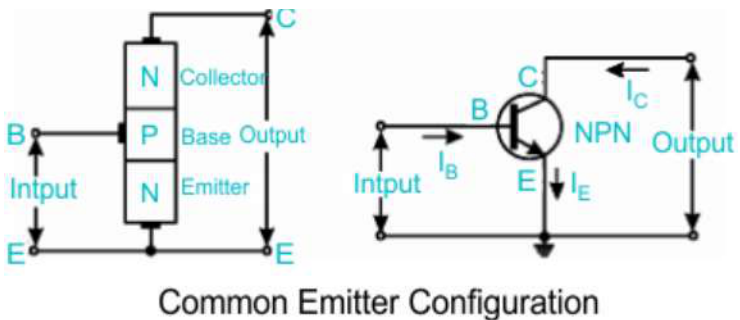


Figure 27: Circuits used to measure the CE characteristics.

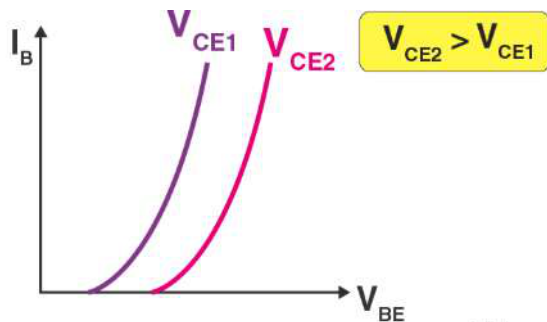
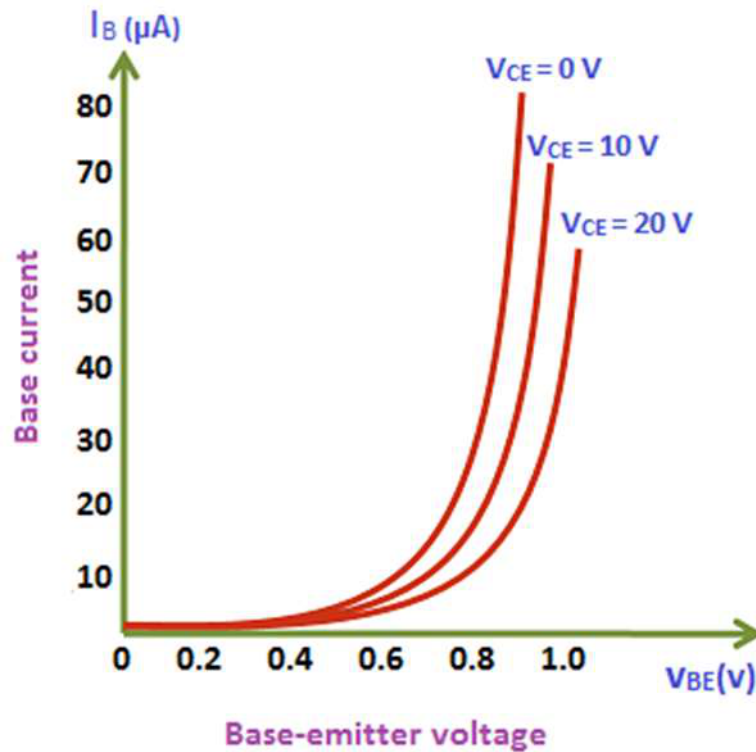


Figure 28: Input characteristics of the CE configuration.



I/P characteristics CE configuration

Figure 29: Input characteristic curves of the CE configuration at different V_{CE} values.

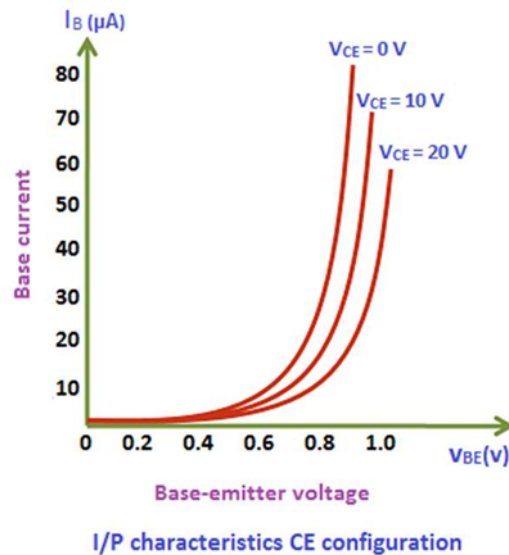


Figure 30: Input characteristics of the CE configuration.

Because the base-emitter junction is forward biased, a small increase in V_{BE} causes a significant increase in I_B , and the input curve resembles a forward-biased PN junction diode. When V_{CE} increases, the reverse bias across the collector-base junction increases, reducing the effective base width (Early effect) and causing a slight shift in the input curves.

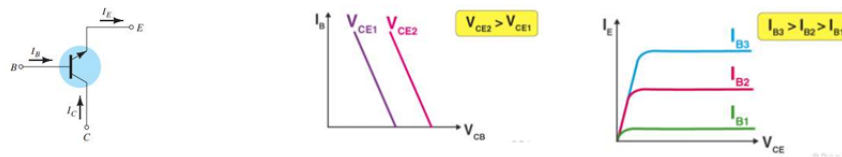
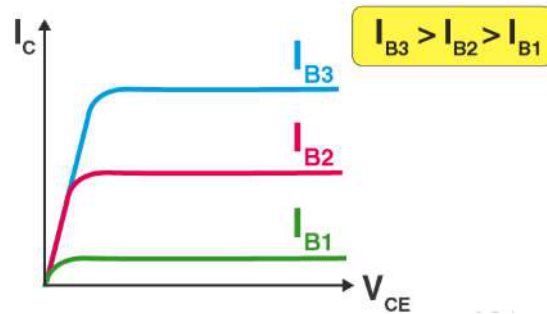
Figure 31: Effect of V_{BE} and V_{CE} on the transistor currents.

Figure 32: Output characteristics of the CE configuration.

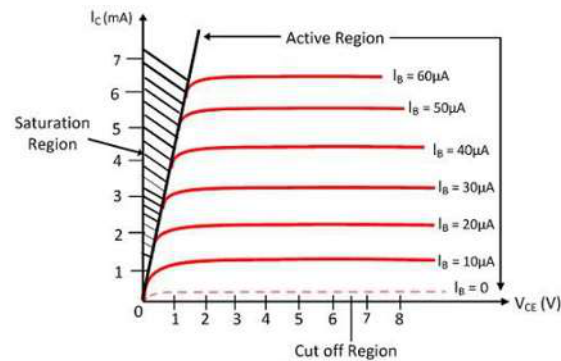


Figure 33: CE output characteristics with the saturation and active regions.

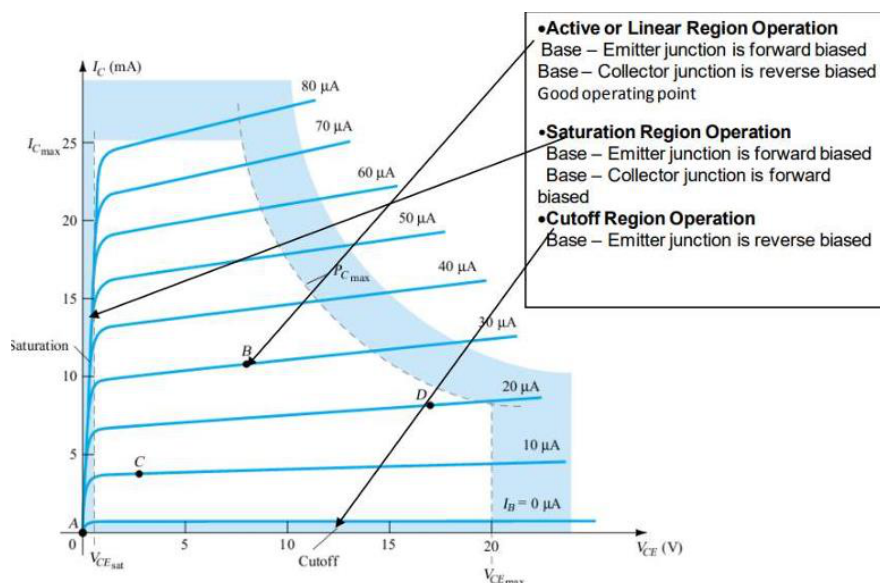


Figure 34: CE output characteristics with all three regions of operation marked.

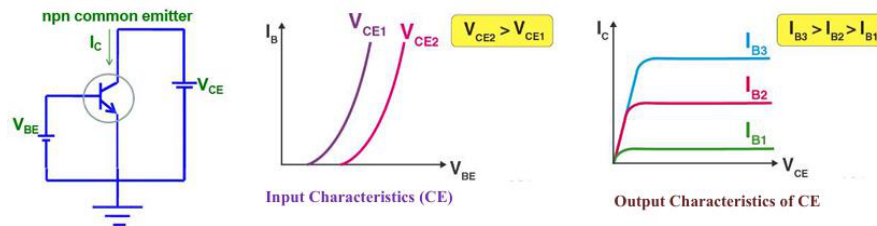


Figure 35: CE output characteristics.

$$\alpha = \frac{\beta}{1 + \beta}$$

Figure 36: CE output characteristics (simplified view).

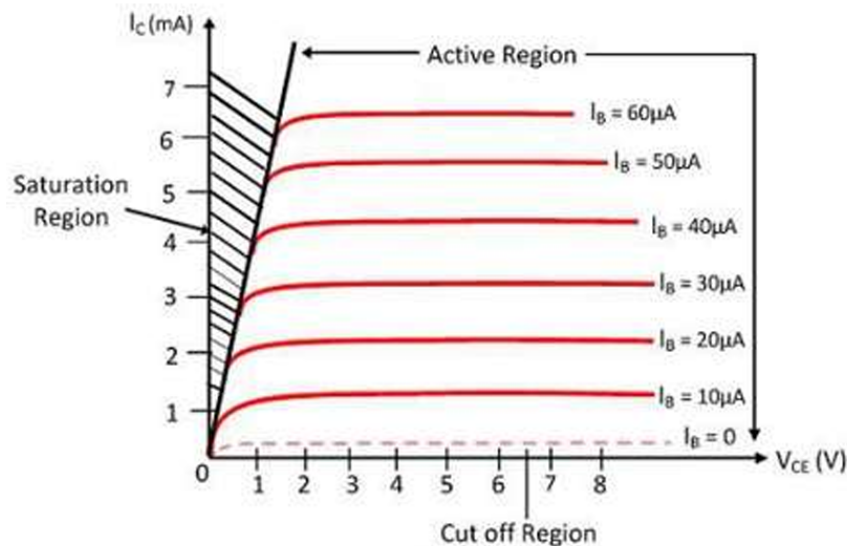


Figure 37: CE output characteristic curves for different base currents.

- **Active region** — the BE junction is forward biased and the CB junction is reverse biased. I_C remains nearly constant over a wide range of V_{CE} and depends mainly on I_B and β .
- **Saturation region** — at very low V_{CE} , both junctions become forward biased and I_C increases rapidly with voltage.
- **Cut-off region** — when $I_B = 0$, only a small leakage current (I_{CEO}) flows through the collector.

$$I_C = \beta \times I_B + I_{CEO}$$

4.3 Common Collector (CC) Configuration Characteristics

In the CC configuration, the collector terminal is common to both input and output.

- **Input characteristics** — the variation of I_B with V_{CB} , keeping V_{CE} constant.
- **Output characteristics** — the variation of I_E with V_{CE} , keeping I_B constant.
- The output characteristics are similar in shape to the CE configuration, but I_E is the output current instead of I_C .

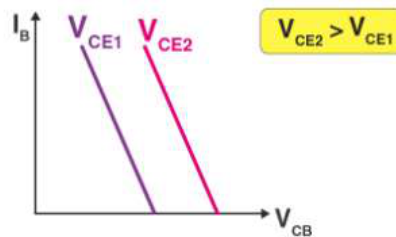


Figure 38: Input characteristics of the CC configuration.

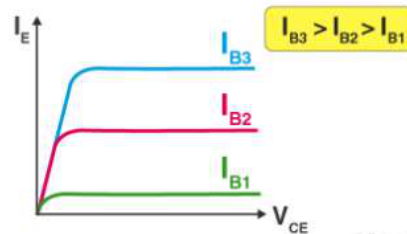


Figure 39: Output characteristics of the CC configuration.

5. Current Gains and Leakage Currents

5.1 Current Gain in Common Base Configuration (alpha)

I_C = Collector current

I_B = Base current

$\beta \gg 1$ (typically 20 – 300)

Figure 40: The three transistor currents in an NPN transistor.

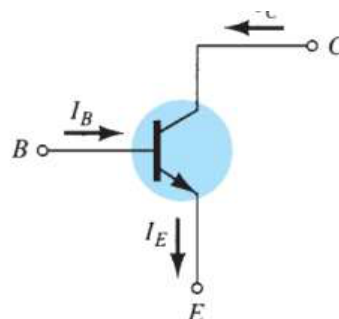


Figure 41: The three transistor currents in a PNP transistor.

$$\alpha = \frac{I_C}{I_E}$$

I_C = Collector current

I_E = Emitter current

$\alpha < 1$ (typically 0.95 – 0.99)

$$I_E = I_C + I_B$$

Figure 42: Current relations and the definition of alpha.

The current transfer ratio **alpha** in the CB configuration is the ratio of the collector current to the emitter current:

$$\alpha = I_C / I_E \quad (\alpha < 1, \text{ typically } 0.95 \text{ to } 0.99)$$

$$\alpha = \frac{I_C}{I_E}$$

I_C = Collector current

I_E = Emitter current

$\alpha < 1$ (typically 0.95 – 0.99)

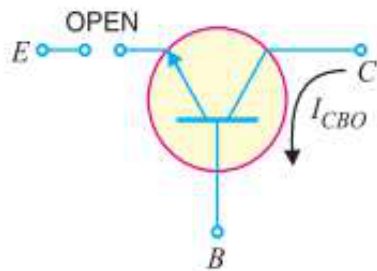
$$I_E = I_C + I_B$$

Figure 43: Definition of the current amplification factor alpha.

CB Leakage Current (I_{CBO})

Leakage current is the small unwanted current that flows in a transistor even when the input junction is not forward biased — mainly due to minority carriers — and it increases with temperature.

- I_{CBO} is the collector-base leakage current. It flows when the emitter is open-circuited and the CB junction is reverse biased.
- When $I_E = 0$, $I_C = I_{CBO}$ — the current caused by minority carriers crossing the PN junction.
- I_{CBO} increases with temperature.

Figure 44: Circuit showing the collector-base leakage current I_{CBO} (emitter open).

$$\begin{aligned} I_C &= \alpha I_E + I_{CBO} \\ I_C &= \alpha I_E + I_{CBO} = \alpha (I_B + I_C) + I_{CBO} \\ I_C (1 - \alpha) &= \alpha I_B + I_{CBO} \\ I_C &= \frac{\alpha}{1 - \alpha} I_B + \frac{1}{1 - \alpha} I_{CBO} \\ I_C &= \beta I_B + I_{CEO} \\ I_{CEO} &= \frac{1}{1 - \alpha} I_{CBO} = (\beta + 1) I_{CBO} \end{aligned}$$

Figure 45: Collector current including the leakage term.

$$I_C = \alpha \times I_E + I_{CBO}$$

5.2 Current Gain in Common Emitter Configuration (beta)

Current gain **beta** is the ratio of the collector current to the base current in the CE configuration:

$$\text{beta} = I_C / I_B \quad (\text{typically } 20 \text{ to } 500)$$

CE Leakage Current (I_{CEO})

I_{CEO} is the collector-emitter leakage current. It flows when the **base is open-circuited** and the CB junction is reverse biased (CE configuration).

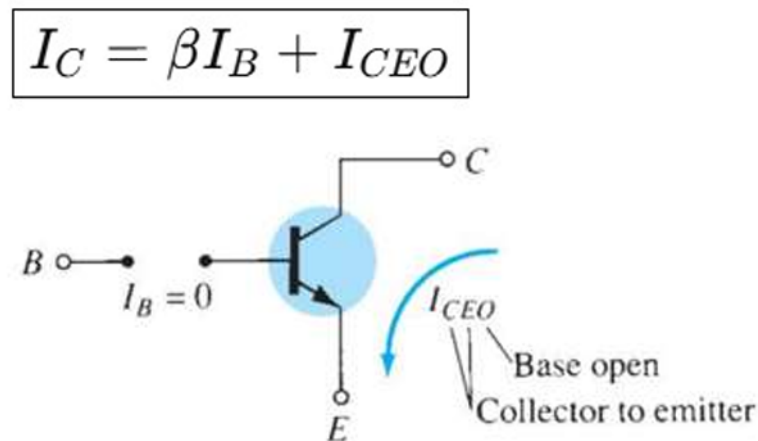


Figure 46: Circuit showing the collector-emitter leakage current I_{CEO} (base open).

$$I_C = \beta I_B + I_{CEO}$$

Figure 47: Collector current including the leakage term (CE configuration).

$$I_C = \text{beta} \times I_B + I_{CEO}$$

$$I_{CEO} = \frac{1}{1 - \alpha} I_{CBO} = (\beta + 1) I_{CBO}$$

Figure 48: Relations between I_{CEO} , beta and the currents.

5.3 Relations Between the Gains and the Leakage Currents

Derivation — relation between I_{CEO} and I_{CBO} : starting from $I_C = \alpha I_E + I_{CBO}$ and substituting $I_E = I_B + I_C$:

$$I_C (1 - \alpha) = \alpha I_B + I_{CBO}$$

$$I_C = [\alpha / (1 - \alpha)] I_B + [1 / (1 - \alpha)] I_{CBO}$$

$$I_C = \text{beta} I_B + I_{CEO}$$

$$I_{CEO} = [1 / (1 - \alpha)] I_{CBO} = (\text{beta} + 1) \times I_{CBO}$$

$$\left[\because \frac{1}{1-\alpha} = \beta + 1 \right]$$

Figure 49: Derivation of the relation between beta and alpha.

$$\begin{aligned} I_C &= \alpha I_E + I_{CBO} \\ I_C &= \alpha I_E + I_{CBO} = \alpha (I_B + I_C) + I_{CBO} \end{aligned}$$

Figure 50: Relation: alpha from beta.

$$\begin{aligned} I_E &= I_C + I_B \\ \alpha &= \frac{I_C}{I_E}, \quad \beta = \frac{I_C}{I_B} \\ \Rightarrow \beta &= \frac{I_C}{I_E - I_C} = \frac{\alpha I_E}{I_E(1-\alpha)} \\ \boxed{\beta} &= \boxed{\frac{\alpha}{1-\alpha}} \end{aligned}$$

$$\boxed{\alpha} = \boxed{\frac{\beta}{1+\beta}}$$

$$\left[\because \frac{1}{1-\alpha} = \beta + 1 \right]$$

Figure 51: The alpha-beta relations and the leakage-current relation.

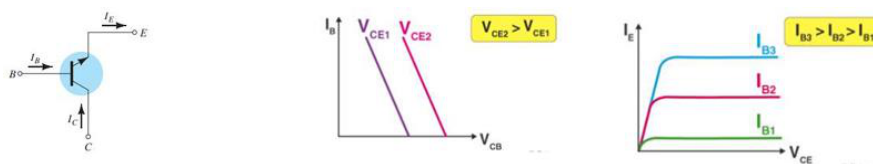


Figure 52: Collector-current equation in terms of alpha.

$$I_C = \frac{\alpha}{1-\alpha} I_B + \frac{1}{1-\alpha} I_{CBO}$$

Figure 53: Collector current expressed using alpha.

$$\begin{aligned}
 I_C &= \alpha I_E + I_{CBO} \\
 I_C &= \alpha I_E + I_{CBO} = \alpha (I_B + I_C) + I_{CBO} \\
 I_C (1 - \alpha) &= \alpha I_B + I_{CBO} \\
 I_C &= \frac{\alpha}{1 - \alpha} I_B + \frac{1}{1 - \alpha} I_{CBO} \\
 I_C &= \beta I_B + I_{CEO} \\
 I_{CEO} &= \frac{1}{1 - \alpha} I_{CBO} = (\beta + 1) I_{CBO}
 \end{aligned}$$

Figure 54: Step-by-step derivation of the current relations.

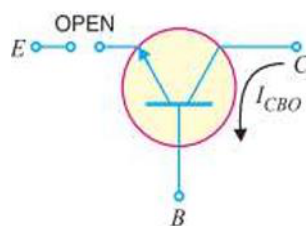
$$\begin{aligned}
 I_E &= I_C + I_B \\
 \alpha &= \frac{I_C}{I_E}, \quad \beta = \frac{I_C}{I_B} \\
 \Rightarrow \beta &= \frac{I_C}{I_E - I_C} = \frac{\alpha I_E}{I_E(1 - \alpha)} \\
 \beta &= \frac{\alpha}{1 - \alpha} \\
 \alpha &= \frac{\beta}{1 + \beta} \\
 \left[\because \frac{1}{1 - \alpha} = \beta + 1 \right]
 \end{aligned}$$

Figure 55: Summary of the alpha and beta relations.

Relation between alpha and beta:

$$\text{beta} = \text{alpha} / (1 - \text{alpha}) \quad \text{alpha} = \text{beta} / (1 + \text{beta})$$

$$\text{Note: } 1 / (1 - \text{alpha}) = \text{beta} + 1$$



$$I_C = \alpha I_E + I_{CBO}$$

Figure 56: Currents marked on the CE circuit.

5.4 Current Gains Summary Table

Parameter	Symbol	Formula	Typical Value
KCL at transistor	$I_E = I_B + I_C$	Valid for NPN and PNP	—
CB current gain	alpha	$\alpha = I_C / I_E$	0.95 to 0.99 (< 1)
CE current gain	beta	$\beta = I_C / I_B$	20 to 500
Beta from alpha	beta	$\beta = \alpha / (1 - \alpha)$	—
Alpha from beta	alpha	$\alpha = \beta / (1 + \beta)$	—
CE leakage current	I_{CEO}	$I_{CEO} = (1/(1-\alpha)) I_{CBO} = (\beta+1) I_{CBO}$	Much larger than I_{CBO}

5.5 Solved Numerical Examples on BJT Currents

These are the standard numerical types asked from this topic — all use only $I_E = I_B + I_C$, $\alpha = I_C / I_E$, $\beta = I_C / I_B$ and the alpha-beta relations.

Example 1: Finding the Base Current

Problem: In a common base connection, $I_E = 1$ mA and $I_C = 0.95$ mA. Calculate the value of I_B .

Solution:

$$I_E = I_B + I_C$$

$$1 = I_B + 0.95 \rightarrow I_B = 1 - 0.95 = 0.05 \text{ mA}$$

$$I_E = I_B + I_C$$

$$1 = I_B + 0.95$$

$$I_B = 1 - 0.95 = 0.05 \text{ mA}$$

Figure 57: Base-current calculation.

Example 2: Finding alpha

Problem: In a common base connection, $I_C = 0.95$ mA and $I_B = 0.05$ mA. Find the value of alpha.

Solution:

$$I_E = I_B + I_C = 0.05 + 0.95 = 1 \text{ mA}$$

$$\text{Current amplification factor: } \alpha = I_C / I_E = 0.95 / 1 = 0.95$$

$$\text{We know } I_E = I_B + I_C = 0.05 + 0.95 = 1 \text{ mA}$$

$$\therefore \text{ Current amplification factor, } \alpha = \frac{I_C}{I_E} = \frac{0.95}{1} = 0.95$$

Figure 58: Calculation of the current amplification factor.

Example 3: Finding the Base Current from alpha

Problem: In a common base connection, the current amplification factor is 0.9. If the emitter current is 1 mA, determine the value of the base current.

Solution:

Here $\alpha = 0.9$ and $I_E = 1$ mA.

$$I_C = \alpha \times I_E = 0.9 \times 1 = 0.9 \text{ mA}$$

$$I_B = I_E - I_C = 1 - 0.9 = 0.1 \text{ mA}$$

$$\text{Here, } \alpha = 0.9, I_E = 1 \text{ mA}$$

$$\text{Now } \alpha = \frac{I_C}{I_E}$$

$$\text{or } I_C = \alpha I_E = 0.9 \times 1 = 0.9 \text{ mA}$$

$$\text{Also } I_E = I_B + I_C$$

$$\therefore \text{Base current, } I_B = I_E - I_C = 1 - 0.9 = 0.1 \text{ mA}$$

Figure 59: Base current from the current amplification factor.

Example 4: Finding beta from alpha

Problem: Find the value of beta if (i) alpha = 0.9, (ii) alpha = 0.98, (iii) alpha = 0.99.

Solution:

Using $\beta = \alpha / (1 - \alpha)$:

$$\text{(i) } \alpha = 0.9: \beta = 0.9 / (1 - 0.9) = 0.9 / 0.1 = 9$$

$$\text{(ii) } \alpha = 0.98: \beta = 0.98 / 0.02 = 49$$

$$\text{(iii) } \alpha = 0.99: \beta = 0.99 / 0.01 = 99$$

Example 5: Leakage Current — CE to CB Arrangement

Problem: The collector leakage current in a transistor is 300 μA in the CE arrangement. If the transistor is now connected in the CB arrangement, what will be the leakage current? Given $\beta = 120$.

Solution:

In the CE arrangement the given leakage current is $I_{\text{CEO}} = 300 \mu\text{A}$, with $\beta = 120$.

Using $I_{\text{CEO}} = (\beta + 1) \times I_{\text{CBO}}$:

$$I_{\text{CBO}} = I_{\text{CEO}} / (\beta + 1) = 300 \mu\text{A} / 121 = 2.48 \mu\text{A}$$

Example 6: Finding I_C Using Both alpha and beta

Problem: Find the alpha rating of the transistor shown in the figure ($\beta = 49$, $I_B = 240 \mu\text{A}$, $I_E = 12 \text{ mA}$). Hence determine the value of I_C using both the alpha and beta ratings.

Solution:

$$\alpha = \beta / (1 + \beta) = 49 / (1 + 49) = 0.98$$

$$\text{Using } \alpha: I_C = \alpha \times I_E = 0.98 \times 12 \text{ mA} = 11.76 \text{ mA}$$

$$\text{Using } \beta: I_C = \beta \times I_B = 49 \times 240 \mu\text{A} = 11.76 \text{ mA}$$

Both methods give the same result: $I_C = 11.76 \text{ mA}$.

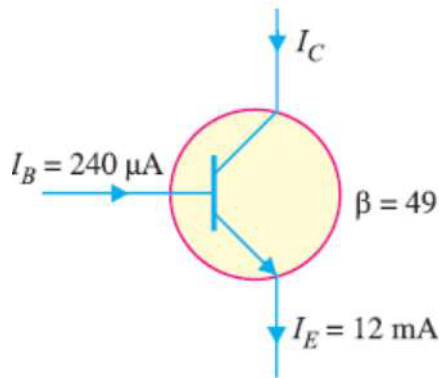


Figure 60: The given transistor with its ratings.

$$\alpha = \frac{\beta}{1 + \beta} = \frac{49}{1 + 49} = 0.98$$

The value of I_C can be found by using either α or β rating as under :

$$I_C = \alpha I_E = 0.98 (12 \text{ mA}) = 11.76 \text{ mA}$$

$$\text{Also } I_C = \beta I_B = 49 (240 \mu\text{A}) = 11.76 \text{ mA}$$

Figure 61: I_C found using both the alpha and beta ratings.

6. BJT Biasing and Load-Line Analysis

6.1 Introduction to Biasing

- The analysis or design of any transistor amplifier has two components: a **DC analysis** and an **AC analysis**.
- Transistor biasing** is the process of setting the transistor's DC operating voltage and current conditions (the Q-point) to the correct level so that any AC input signal can be amplified correctly.
- For the BJT to operate in its linear (active) region: the base-emitter junction must be forward biased AND the base-collector junction must be reverse biased.

The types of transistor biasing are:

- 1 Fixed Bias
- 2 Emitter Bias
- 3 Voltage Divider Bias
- 4 Collector Feedback Bias

6.2 Fixed Bias Method

In the fixed bias method, a single resistor R_B is connected between the base and the DC supply V_{CC} to provide the required base current. The base-emitter junction is forward biased by the base supply, and the collector-emitter circuit is powered by the same supply.

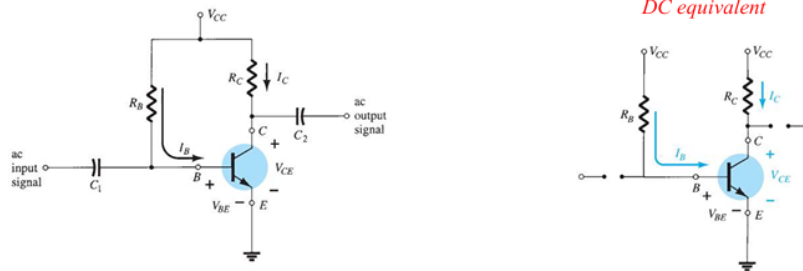


Figure 62: The fixed bias circuit — a single resistor R_B from V_{CC} to the base.

DC analysis of the fixed bias circuit:

Applying KVL to the base-emitter loop:

$$+V_{CC} - I_B R_B - V_{BE} = 0$$

Solving for the base current:

$$I_B = (V_{CC} - V_{BE}) / R_B$$

The collector current is obtained from the base current:

$$I_C = \beta \times I_B$$

Applying KVL to the collector-emitter loop:

$$V_{CE} + I_C R_C - V_{CC} = 0$$

$$V_{CE} = V_{CC} - I_C R_C$$

The voltage at the base with respect to ground can be determined

$$V_B = V_{CC} - I_B R_B$$

$$V_B = V_{BE} + V_E$$

Figure 63: Base-circuit equations used in the DC analysis.

Substituting $I_E \cong I_C$

$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$

$$V_{CE} = V_C - V_E$$

$$V_C = V_{CE} + V_E$$

$$V_E = I_E R_E$$

$$V_C = V_{CC} - I_C R_C$$

Figure 64: Emitter and collector-emitter voltage equations.

The collector-emitter loop equation that defines the load line is

$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$

Choosing $I_C = 0$ mA gives

$$V_{CE} = V_{CC} |_{I_C=0 \text{ mA}}$$

as obtained for the fixed-bias configuration. Choosing $V_{CE} = 0$ V gives

$$I_C = \frac{V_{CC}}{R_C + R_E} |_{V_{CE}=0 \text{ V}}$$

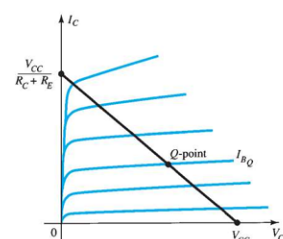


Figure 65: Load-line end-points from the collector-emitter loop.

Substituting $V_{GS} = 0$ V gives

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

$$= I_{DSS} \left(1 - \frac{0}{V_P} \right)^2 = I_{DSS} (1 - 0)^2$$

$$I_D = I_{DSS} \mid_{V_{GS}=0 \text{ V}}$$

V_{GS} versus I_D Using Shockley's Equation

V_{GS}	I_D
0	I_{DSS}
$0.3V_P$	$I_{DSS}/2$
$0.5V_P$	$I_{DSS}/4$
V_P	0 mA

Substituting $V_{GS} = V_P$ yields

$$I_D = I_{DSS} \left(1 - \frac{V_P}{V_P} \right)^2$$

$$= I_{DSS} (1 - 1)^2 = I_{DSS} (0)$$

$$I_D = 0 \text{ A} \mid_{V_{GS}=V_P}$$

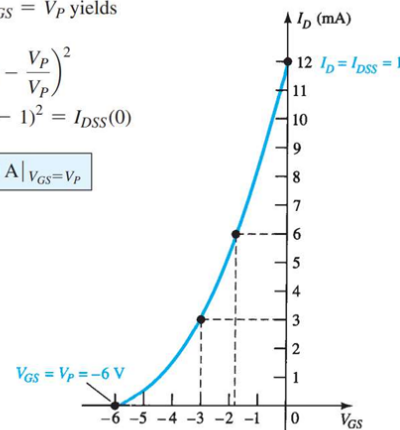


Figure 66: V_{CE} equations with the I_C - V_{CE} graph.

Since the emitter is grounded ($V_E = 0$ V): $V_{CE} = V_C$ and $V_{BE} = V_B$.

6.3 Load-Line Analysis and the Q-Point

- Load-line analysis is a graphical method used to determine the operating point (Q-point) of a transistor on its output characteristics.
- The DC load line shows all possible combinations of I_C and V_{CE} for a given V_{CC} and R_C .
- The load line helps in selecting a proper operating point so that the transistor remains in the active region during signal amplification.
- The load line equation comes from: $V_{CE} = V_{CC} - I_C R_C$

Finding the two end-points of the load line

- When $I_C = 0$ mA: $V_{CE} = V_{CC}$ — this is the voltage intercept on the x-axis ($V_{CE(max)} = V_{CC}$).
- When $V_{CE} = 0$ V: $I_C = V_{CC} / R_C$ — this is the current intercept on the y-axis ($I_{C(max)} = V_{CC} / R_C$).

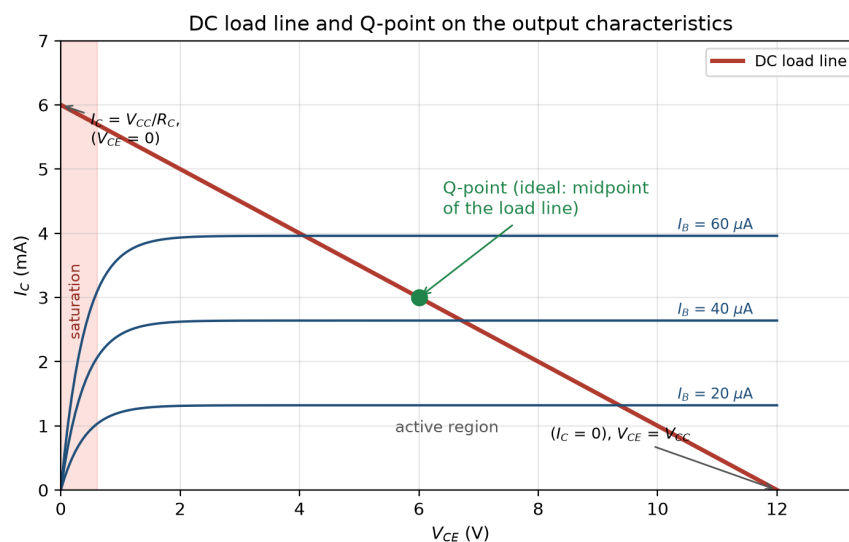


Figure 67: The DC load line drawn on the output characteristics — the Q-point is its intersection with the curve for the actual base current.

Q-Point (Quiescent Point): the point where the DC load line intersects the transistor output characteristic corresponding to a particular base current I_B — this is the DC operating point. An ideal

Q-point is located at the **midpoint of the load line**, so that the output signal can swing symmetrically in both directions without distortion or clipping.

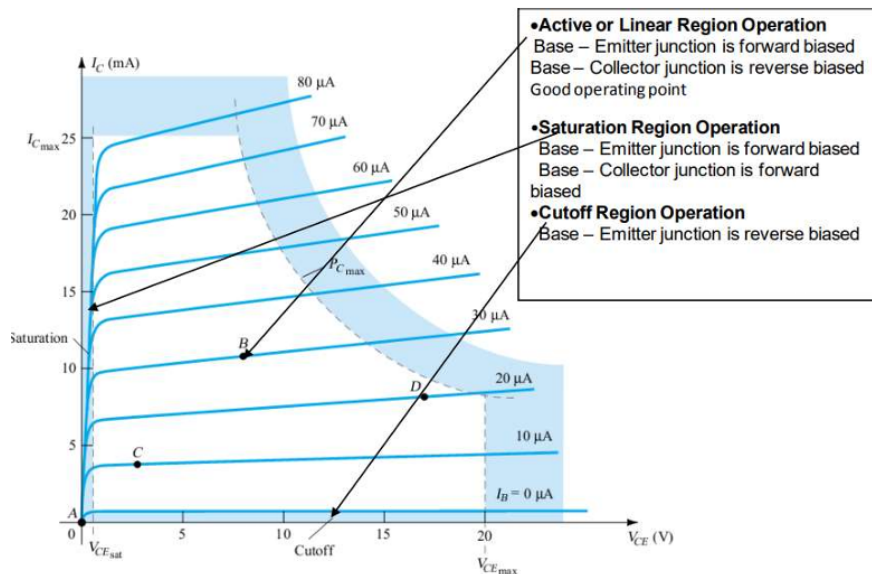
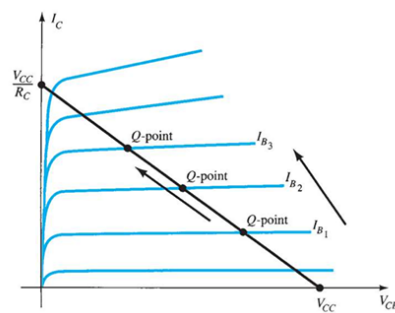
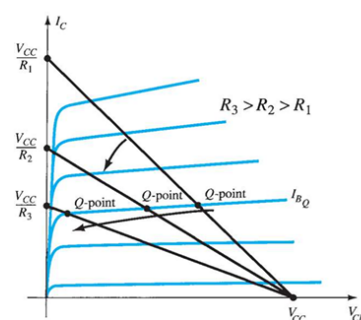


Figure 68: Output characteristics with the operating regions and $V_{CE(sat)}$

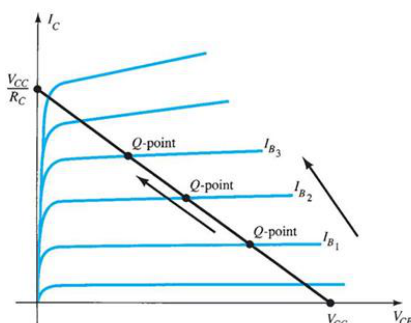


Movement of the Q-point with increasing level of I_B .

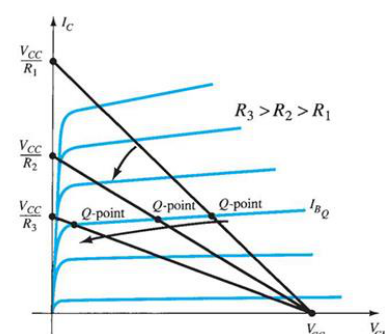


Effect of an increasing level of R_C on the load line and the Q-point.

Figure 69: Movement of the Q-point (Q_1 , Q_2 , Q_3) with different base currents.



Movement of the Q-point with increasing level of I_B .



Effect of an increasing level of R_C on the load line and the Q-point.

Figure 70: Effect of changing R_C and V_{CC} on the load line.

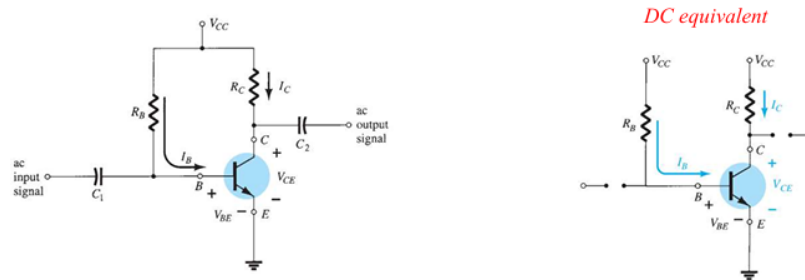


Figure 71: DC equivalent circuit and the load line.

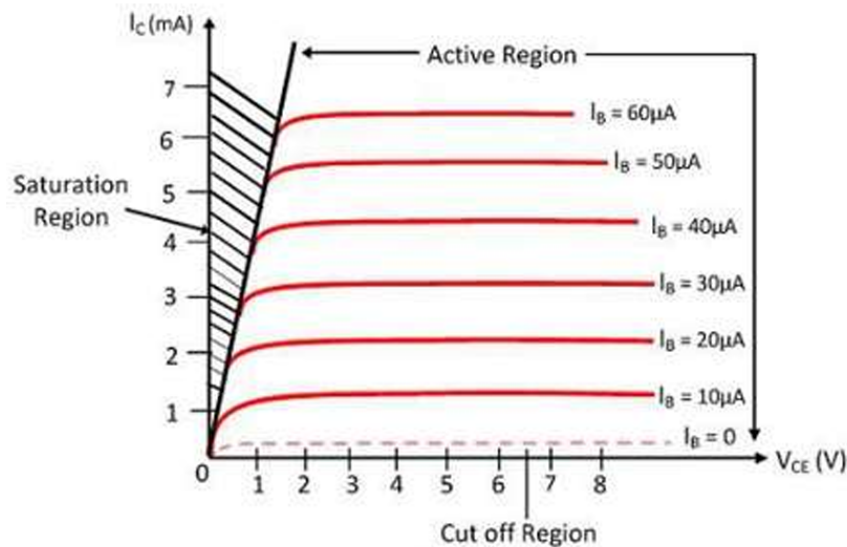


Figure 72: The output characteristics on which the load line is drawn.

6.4 Emitter Bias Method

The fixed bias method is simple and easy to analyse, but it provides **poor stability** because the collector current depends heavily on the transistor's beta. A small change in beta or temperature can cause a large change in the collector current, which may shift the operating point out of the active region.

In the emitter bias method, an **emitter resistor R_E** is added to improve stability. The voltage drop across the emitter resistor increases with an increase in emitter current, which in turn reduces the base-emitter voltage — so the circuit automatically opposes changes in the operating point.

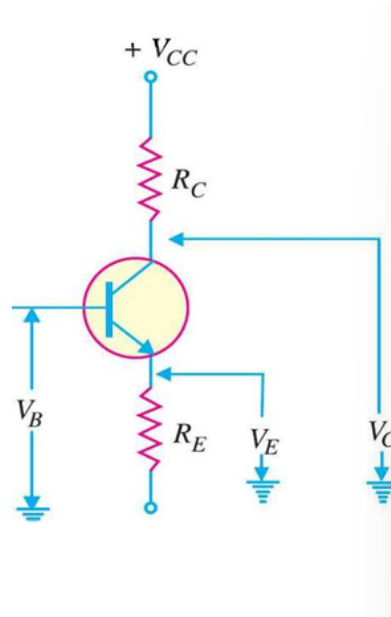


Figure 73: The emitter bias circuit — the emitter resistor R_E improves the stability of the Q-point.

Base-emitter loop: applying KVL,

$$+V_{CC} - I_B R_B - V_{BE} - I_E R_E = 0$$

We know the relation between the base and emitter currents: $I_E = (\beta + 1) I_B$. Substituting:

$$V_{CC} - I_B R_B - V_{BE} - (\beta + 1) I_B R_E = 0$$

Solving for I_B :

$$I_B = (V_{CC} - V_{BE}) / [R_B + (\beta + 1) R_E]$$

Collector-emitter loop: applying KVL,

$$V_{CC} - I_C R_C - V_{CE} - I_E R_E = 0$$

Substituting I_E approximately equal to I_C :

$$V_{CE} = V_{CC} - I_C (R_C + R_E)$$

The addition of the emitter resistor provides **improved stability** — the DC bias currents and voltages remain closer to their set values when outside conditions such as temperature and transistor beta change.

6.5 Voltage Divider Bias

In the previous bias configurations, the Q-point depended on beta, which is temperature sensitive. The **voltage divider bias** arrangement reduces this dependency on beta: two resistors R_1 and R_2 form a voltage divider at the base, so the base voltage is fixed by the divider and barely changes with beta.

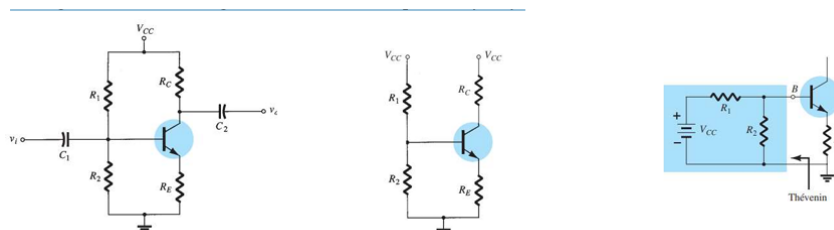


Figure 74: The voltage divider bias circuit (left) and its Thevenin equivalent seen from the base (right).

Analysing the circuit to the left of the base terminal with a Thevenin equivalent network (the voltage source replaced by a short circuit):

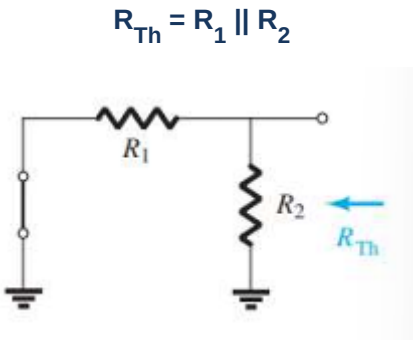


Figure 75: The Thevenin equivalent of the divider network: V_{Th} in series with R_{Th} .

This is the most widely used bias arrangement in practice because of its excellent stability.

6.6 Collector Feedback Bias

In the collector feedback bias, the base resistor R_B is connected from the **collector to the base** instead of from V_{CC} . If the collector current rises, the voltage drop across R_C increases, the collector voltage falls, and the base current is automatically reduced — the circuit therefore **self-stabilises** using only a single supply.

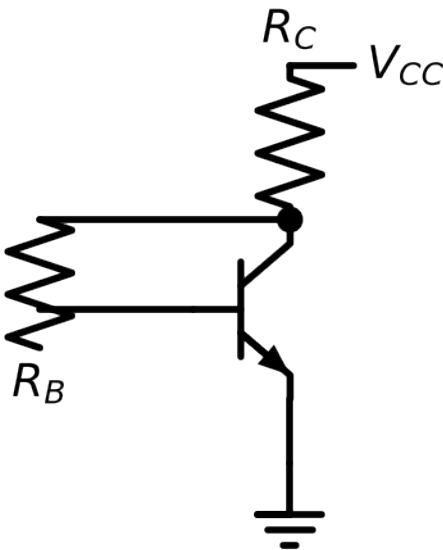


Figure 76: The collector feedback bias circuit — R_B connects the collector to the base.

6.7 Biasing Methods Comparison

Bias Method	Key Feature	Stability	Use Case
Fixed Bias	Single resistor R_B from V_{CC} to base	Poor (Q-point shifts with beta)	Simple circuits, lab demos
Emitter Bias	Emitter resistor R_E added for stability	Better than fixed	Moderate-stability circuits
Voltage Divider Bias	Two resistors (R_1 , R_2) form a voltage divider	Excellent (most stable)	Most widely used in practice
Collector Feedback Bias	Feedback resistor from collector to base	Good (self-stabilising)	Single-supply circuits

7. Junction Field Effect Transistor (JFET)

7.1 Introduction to Field Effect Transistors

- A Field Effect Transistor (FET) is a **voltage-controlled** device — the output characteristics are controlled by the input voltage (not the input current).
- There are two basic types of FETs: the **Junction Field Effect Transistor (JFET)** and the **Metal Oxide Semiconductor FET (MOSFET)**.

7.2 JFET — Definition and Terminals

- A JFET is a three-terminal semiconductor device in which current conduction is by **one type of carrier only** (electrons OR holes) — hence it is a **unipolar** device.
- Terminals: **Source (S)**, **Drain (D)** and **Gate (G)**.
- Current conduction is controlled by an electric field between the gate and the conducting channel.
- Two types: **N-channel JFET** (more common, generally preferred) and **P-channel JFET**.

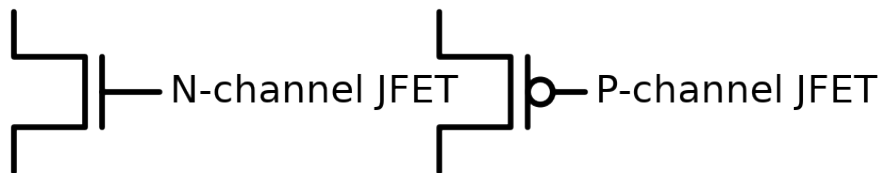


Figure 77: Circuit symbols of the N-channel and P-channel JFET.

7.3 Construction of the JFET (N-Channel)

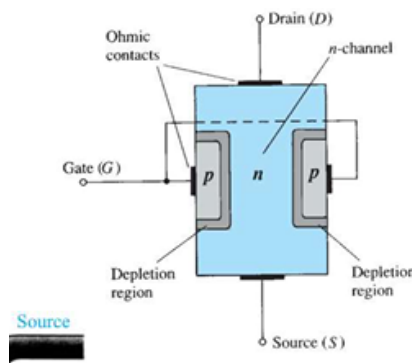


Figure 78: Construction of the N-channel JFET.

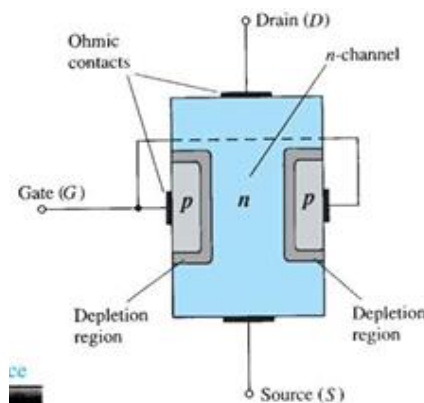


Figure 79: N-channel JFET structure with the depletion regions.

- An N-type silicon bar, referred to as the **channel**, has two smaller pieces of P-type silicon diffused on opposite sides of its middle part, forming two PN junctions.

- The two PN junctions (forming diodes or gates) are connected internally, and a common terminal called the **gate terminal (G)** is brought out.
- Ohmic contacts (direct electrical connections) are made at the two ends of the channel: one lead is the **Source (S)** and the other is the **Drain (D)**.
- **Source:** the terminal through which the majority carriers (electrons for N-channel) enter the channel.
- **Drain:** the terminal through which the majority carriers leave the channel.
- **Channel:** the region between the source and drain, sandwiched between the two gate regions, through which the majority carriers flow.
- **Gate:** controls the width of the channel using a reverse-biased PN junction, and hence controls the drain current.

7.4 Working Principle of the JFET

Consider the N-channel JFET with the gate reverse biased with respect to the source:

- When a voltage V_{DD} is applied between the drain and source terminals with the gate voltage at zero, the two PN junctions at the sides of the bar establish depletion layers.
- Electrons flow from the source to the drain through the channel between the depletion layers.
- The size of the depletion layers determines the width of the channel, and hence the current through the bar.
- For small V_{DD} , the N-type bar acts as a simple semiconductor resistor: the drain current increases linearly with V_{DS} up to the knee point.
- As V_{DD} increases further, the conducting part of the channel begins to constrict more at the drain end, until a voltage is reached at which the channel is **pinched off**.
- Beyond pinch-off, the drain current I_D no longer increases with V_{DS} — it approaches a constant saturation value (I_{DSS} , the drain-source saturation current, when $V_{GS} = 0$ V).

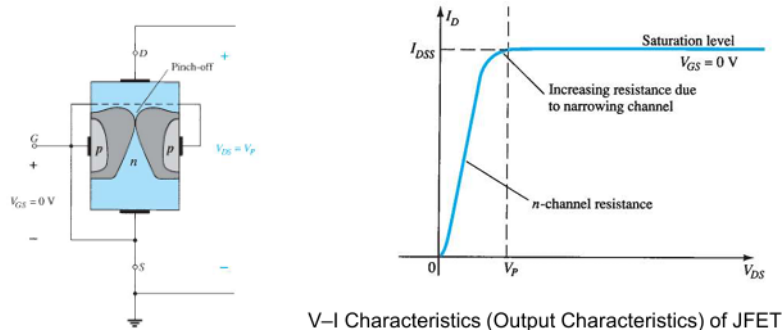


Figure 80: Drain (output) characteristics of the JFET — I_D versus V_{DS} for different V_{GS} values.

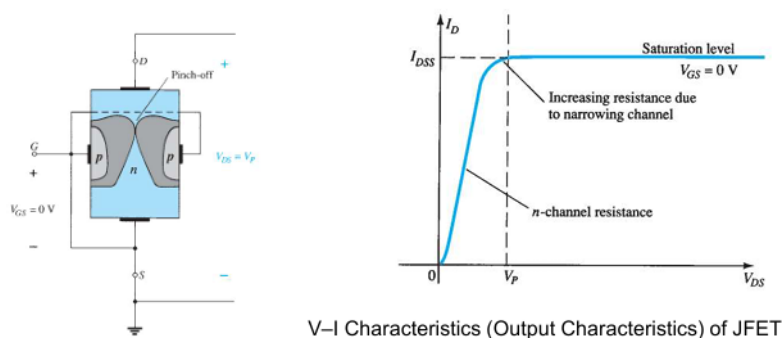


Figure 81: JFET output characteristics showing the ohmic region, the knee and the pinch-off (saturation) region.

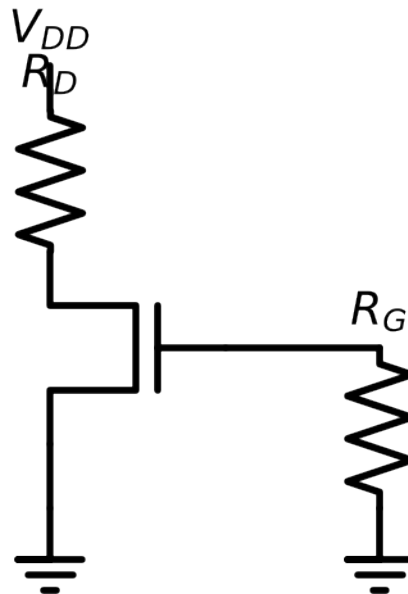


Figure 82: A JFET connected in the common-source arrangement: V_{DD} through R_D to the drain, source grounded, gate through R_G .

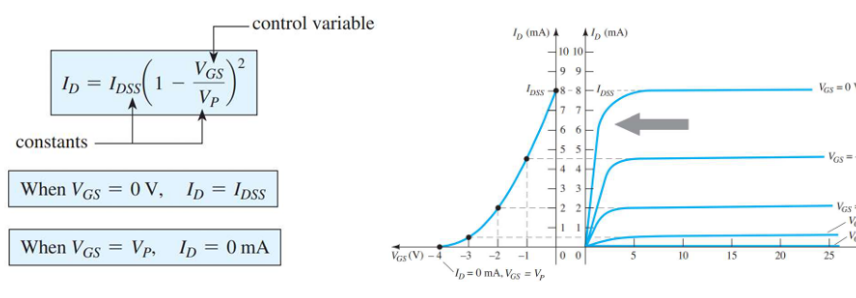


Figure 83: JFET circuit with the current equations.

Effect of the gate-source voltage (V_{GS})

- When a reverse voltage V_{GS} is applied between the gate and source terminals, the width of the depletion layer increases. This reduces the width of the conducting channel, increasing the resistance of the bar — so the drain current **decreases**.
- On the other hand, when the reverse bias on the gate is decreased, the depletion layer narrows, the channel widens, and the drain current **increases**.

7.5 Pinch-Off Voltage and Shockley's Equation

The **pinch-off voltage (V_P)** is the value of V_{DS} at which the channel is completely pinched off (all the free charges are removed from the channel) and the drain current I_D attains a constant value. Beyond pinch-off, I_D saturates.

The drain current in the pinch-off (saturation) region is given by **Shockley's equation** for the JFET:

$$I_D = I_{DSS} \times \left(1 - V_{GS} / V_P\right)^2$$

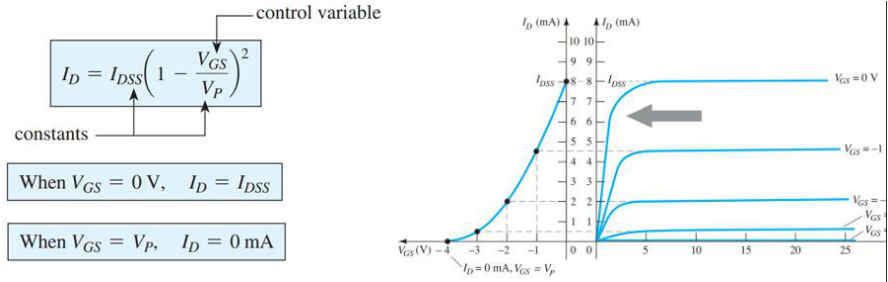


Figure 84: Shockley's equation for the JFET and the related relations.

Symbol	Meaning
I_D	Drain current at the operating point
I_{DSS}	Drain-to-source saturation current (maximum I_D when $V_{GS} = 0\text{ V}$)
V_{GS}	Gate-to-source voltage (the control input)
V_P	Pinch-off voltage (at which $I_D = 0\text{ mA}$)

7.6 Transfer Characteristics of the JFET

For the BJT, the output current I_C and the input controlling current I_B are related by beta: $I_C = f(I_B) = \text{beta} \times I_B$ (beta assumed constant). Similarly, for the JFET the relationship between I_D and V_{GS} is defined by Shockley's equation.

The transfer characteristics show the relationship between the drain current I_D and the gate-source voltage V_{GS} (the control variable):

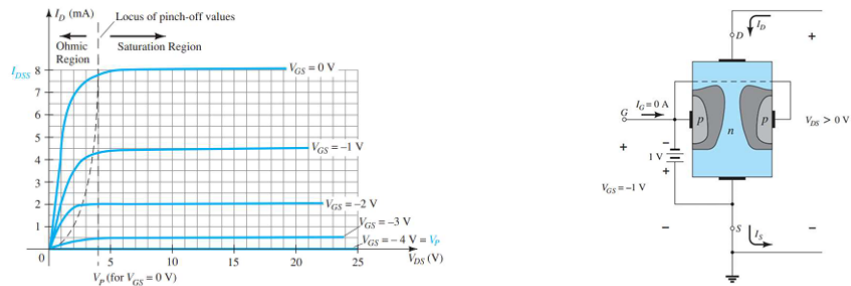


Figure 85: Transfer characteristics of the JFET — I_D versus V_{GS} .

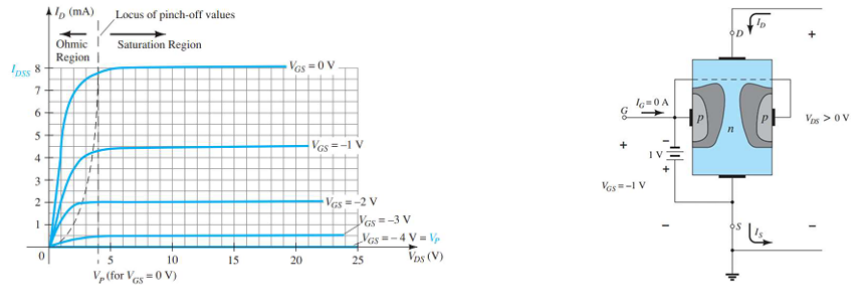


Figure 86: Transfer characteristic derived from Shockley's equation.

Substituting $V_{GS} = 0 \text{ V}$ gives

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2 \\ = I_{DSS} \left(1 - \frac{0}{V_P} \right)^2 = I_{DSS} (1 - 0)^2$$

$$I_D = I_{DSS} \mid_{V_{GS}=0 \text{ V}}$$

V_{GS} versus I_D Using Shockley's Equation

V_{GS}	I_D
0	I_{DSS}
$0.3V_P$	$I_{DSS}/2$
$0.5V_P$	$I_{DSS}/4$
V_P	0 mA

Substituting $V_{GS} = V_P$ yields

$$I_D = I_{DSS} \left(1 - \frac{V_P}{V_P} \right)^2 \\ = I_{DSS} (1 - 1)^2 = I_{DSS} (0)$$

$$I_D = 0 \text{ A} \mid_{V_{GS}=V_P}$$

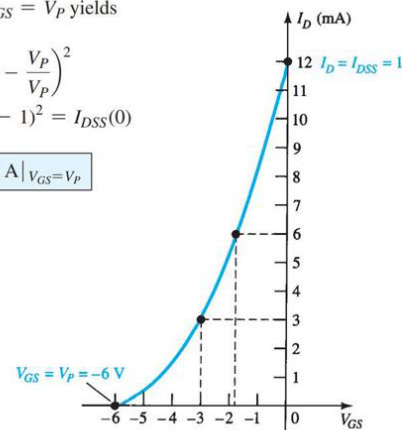


Figure 87: The I_D – V_{GS} relationship of the JFET.

Key points from Shockley's JFET equation

- When $V_{GS} = 0 \text{ V}$: $I_D = I_{DSS} (1 - 0)^2 = I_{DSS}$ — the maximum drain current.
- When $V_{GS} = V_P$: $I_D = I_{DSS} (1 - 1)^2 = 0 \text{ mA}$ — the channel is fully pinched off.

Transfer characteristic reference table

V_{GS}	I_D (drain current)	Notes
0 V	I_{DSS}	Maximum drain current; channel fully open
$0.3 V_P$	$I_{DSS} / 2$	Half of the maximum current
$0.5 V_P$	$I_{DSS} / 4$	Quarter of the maximum current
V_P (pinch-off)	0 mA	Channel fully pinched off; no drain current

7.7 Comparison: BJT vs. JFET

Parameter	BJT	JFET
Control type	Current controlled (I_B controls I_C)	Voltage controlled (V_{GS} controls I_D)
Terminals	Base, Emitter, Collector	Gate, Source, Drain
Input impedance	Low to medium (k-ohm range)	Very high (M-ohm range)
Carriers	Both electrons and holes (bipolar)	One type only (unipolar)
Noise	Higher	Lower
Gain	High current and voltage gain	Moderate voltage gain
Control mechanism	Base current	Gate voltage (reverse biased)
Key equation	$I_C = \beta I_B + I_{CEO}$	$I_D = I_{DSS} (1 - V_{GS}/V_P)^2$
Applications	Amplifiers, switching, logic	Amplifiers, switches, analog circuits

8. Quick Reference — Key Formulae and Definitions

8.1 BJT Key Formulae

Quantity	Formula	Notes
KCL at transistor	$I_E = I_B + I_C$	Valid for both NPN and PNP
CB current gain	$\alpha = I_C / I_E$	0.95–0.99 (< 1)
CE current gain	$\beta = I_C / I_B$	20–500
Beta from alpha	$\beta = \alpha / (1 - \alpha)$	—
Alpha from beta	$\alpha = \beta / (1 + \beta)$	—
Collector current (CE)	$I_C = \beta I_B + I_{CEO}$	Includes the leakage term
CE leakage	$I_{CEO} = (\beta + 1) \times I_{CBO}$	Much larger than I_{CBO}
Fixed bias: I_B	$I_B = (V_{CC} - V_{BE}) / R_B$	V_{BE} is about 0.7 V for Si
Fixed bias: V_{CE}	$V_{CE} = V_{CC} - I_C R_C$	Active region: $V_{CE} > V_{CE(sat)}$
Emitter bias: I_B	$I_B = (V_{CC} - V_{BE}) / [R_B + (\beta + 1) R_E]$	Emitter resistor improves stability
Load line intercepts	$I_{C(max)} = V_{CC} / R_C$; $V_{CE(max)} = V_{CC}$	End-points of the DC load line

8.2 JFET Key Formulae

Quantity	Formula	Notes
JFET drain current (Shockley)	$I_D = I_{DSS} (1 - V_{GS} / V_P)^2$	Saturation / pinch-off region
At $V_{GS} = 0$	$I_D = I_{DSS}$	Maximum drain current
At $V_{GS} = V_P$	$I_D = 0$ mA	Channel fully pinched off
At $V_{GS} = 0.3 V_P$	$I_D = I_{DSS} / 2$	Half of the maximum current
At $V_{GS} = 0.5 V_P$	$I_D = I_{DSS} / 4$	Quarter of the maximum current

8.3 Key Definitions at a Glance

Term	Definition
BJT	Bipolar Junction Transistor — a current-controlled, three-terminal device using both electron and hole conduction.
JFET	Junction Field Effect Transistor — a voltage-controlled, three-terminal, unipolar device.
alpha	CB current gain = I_C/I_E ; always less than 1.
beta	CE current gain = I_C/I_B ; typically 20–500.
Active region	BJT operating region for linear amplification; EB junction forward biased, CB junction reverse biased.
Saturation region	Both junctions forward biased; the transistor acts as a closed switch.
Cut-off region	Both junctions reverse biased; the transistor acts as an open switch.
Q-Point	Quiescent (DC operating) point — the intersection of the load line with the transistor characteristic for a given I_B .
Load line	Graphical representation of all possible I_C vs V_{CE} combinations for given V_{CC} and R_C .
Pinch-off voltage (V_P)	The drain voltage at which the JFET channel is completely closed and I_D becomes constant.
I_{DSS}	Drain-source saturation current — the maximum drain current when $V_{GS} = 0$ V.
Early effect	Base-width modulation — as V_{CB} increases, the depletion region widens and the effective base width reduces.
I_{CBO}	CB leakage current with the emitter open, caused by thermally generated minority carriers.
I_{CEO}	CE leakage current with the base open; $I_{CEO} = (\text{beta} + 1) I_{CBO}$.